Experiment M11 bis: Torsional pendulum, moment of inertia and Steiner's theorem

[redacted]

Group B2.2

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Abstract

This experiment analyzes a torsional pendulum to determine the inertial properties of rigid bodies. First, the spring's torsion constant is derived by verifying the linear relationship between applied torque and angular displacement. Subsequently, the moments of inertia for various geometric shapes are obtained experimentally via oscillation periods and compared with theoretical predictions. Finally, the Parallel Axis Theorem (Steiner's Theorem) is verified by examining the dependence of the oscillation period on the distance between the rotation axis and the center of mass

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Avoid categorical statements:
'robust statistical confirmation'
'fully explained by uncertainties'
'validate theoretical models'

1 Results and Discussion

1.1 Part A: Determination of the Torsion Constant (D)

To determine the torsion constant (D) of the spring, the angular displacement (θ) was measured as a function of the applied torque (τ). The torque was calculated using the force measured by the dynamometer and the radius of the support disc ($R = 0.14$ m).

The experimental data were fitted using a linear regression model based on Hooke's Law for torsion:

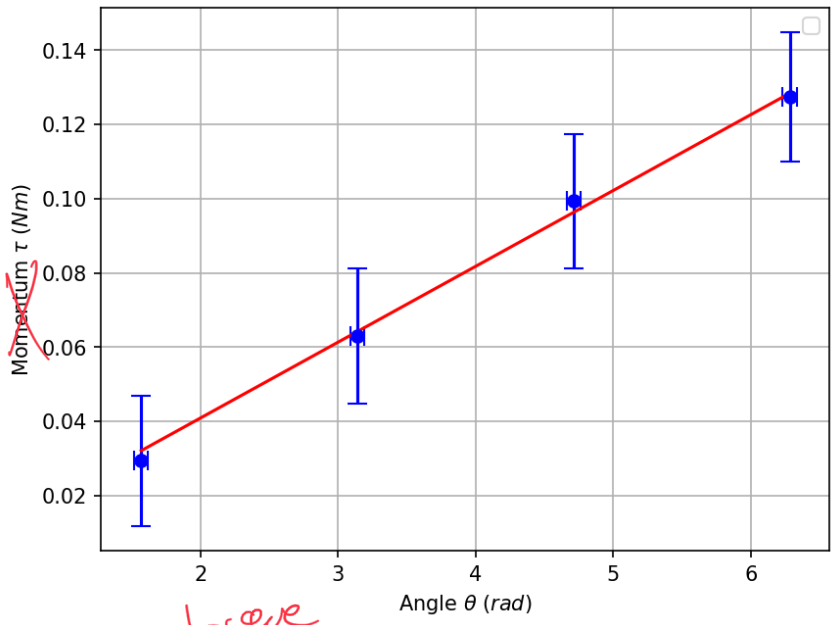
$$\tau = D \cdot \theta \tag{1}$$

Figure 1 shows the linear relationship between the applied torque and the rotation angle. To evaluate the goodness of the fit, the Pearson correlation coefficient and the reduced chi-squared statistic (χ^2_{red}) were analyzed.

The high Pearson coefficient ($R^2 = 0.9987097694$) indicates a strong linear dependence. Furthermore, the reduced chi-squared value of $\chi^2_{red} = 0.9654706217$ is remarkably close to unity. A value of $\chi^2_{red} \approx 1$ signifies that the scatter of the data points around the fitted line is fully explained by the experimental uncertainties, implying that the linear model is appropriate and that there are no significant non-linear effects or systematic errors in the measurement range. This provides robust statistical confirmation of Hooke's Law for this system.

no need for that many digits

one



torque

torque

Figure 1: Moment of the system (τ) in function of the Angle of rotation (θ).

The torsion constant D corresponds to the slope of the fitted line. The result obtained is:

$$D = 0.02104 \pm 0.00078\,\mathrm{N \cdot m/rad} \tag{2}$$

1.2 Part B: Determination of Moments of Inertia

The moments of inertia for various geometric bodies were determined experimentally by measuring their oscillation period (T) and applying the period equation:

$$I_{exp} = \frac{T^2 D}{4\pi^2} \tag{3}$$

These values were compared with the theoretical moments of inertia (I_{theo}), calculated using the mass and geometric dimensions of each body.

Table 1 summarizes the results, including the experimental uncertainties and the relative discrepancy between theoretical and experimental values.

Table 1: Comparison between Theoretical and Experimental Moments of Inertia.

Object	**Mass (kg)**	I_{theo} **(kg · m^2)**	I_{exp} **(kg · m^2)**	**Discrepancy (%)**
Graduated Disc	0.2442	0.002750 ± 0.000011	0.00226 ± 0.00035	18%
White Disc	0.2651	0.0015500 ± 0.0000083	0.00151 ± 0.00023	2.6%
Rod	0.1710	0.0051300 ± 0.0000059	0.00371 ± 0.00057	28%
Sphere	0.8580	0.001500 ± 0.000051	0.00174 ± 0.00027	16%
Hollow Cylinder	0.3389	0.0007400 ± 0.0000082	0.00101 ± 0.00016	36%

The experimental results show a reasonable agreement with the theoretical predictions. Notably, the White Disc shows excellent agreement with a discrepancy of only 2.6%, falling well within the experimental uncertainty. The Sphere and Graduated Disc also show acceptable agreement (16% and 18% respectively). Note that uncertainties regarding weight have been omitted due to the precision (0.001 g) of the electric scale.

Some larger discrepancies persist for the Rod (28%) and Hollow Cylinder (36%). These deviations might be attributed to geometric approximations, such as the rod's axis not being perfectly centered, due to a small deviation on the plate that we use to measure the object, the small deviation could have caused more error in the moment of the rod, since their mass distribution is more spreaded on the horizontal plane, thus causing that in the ends the small deviation of the angle becames more signficant, causing more error, or the cylinder's density distribution, or minor residual errors in the period measurement. Overall, the values obtained validate the theoretical models for moments of inertia.

1.3 Part C: Calculation of Moment of Inertia using Steiner's Theorem

In this section, we verify the Parallel Axis Theorem (Steiner's Theorem) by analyzing the dependence of the oscillation period T on the distance a between the rotation axis and the center of mass. Experimental measurements were taken for distances $a = 0$, 0.03, 0.06, 0.09, and 0.12 meters. According to the theoretical model derived in equation (9), the relationship is linearized as $T^2 = \frac{4\pi^2}{D} I_{cm} + \frac{4\pi^2 M}{D} a^2$.

The graphical representation of T^2 versus a^2 is presented below in 2. The data points show a clear linear trend, which supports the theoretical prediction.

The reduced chi-squared value obtained was 0.9492372829 which is close to 1 and it indicates that the linear model provides an excellent description of the observed data and that the experimental uncertainties have been estimated correctly. We can conclude that the reduced chi-squared indicated that the fit is statistically significant and the discrepancies are purely random.

To quantitatively verify the theory, we compare the Experimental Slope obtained from the fit with the Theoretical Factor calculated using the mass M measured on the balance and the torsion constant D.

The Theoretical Factor is calculated as:

$$A_{theo} = \frac{4\pi^2 M_{disc}}{D} = 459.08\,\mathrm{s}^2/\mathrm{m}^2 \tag{4}$$

On the other hand, the experimental slope obtained directly from the regression is:

$$A_{exp} = 411 \pm 40\,\mathrm{s}^2/\mathrm{m}^2 \tag{5}$$

Comparing both values, we observe a percentage discrepancy of:

$$\varepsilon = \left| \frac{A_{exp} - A_{theo}}{A_{theo}} \right| \times 100 = 10.48\% \tag{6}$$

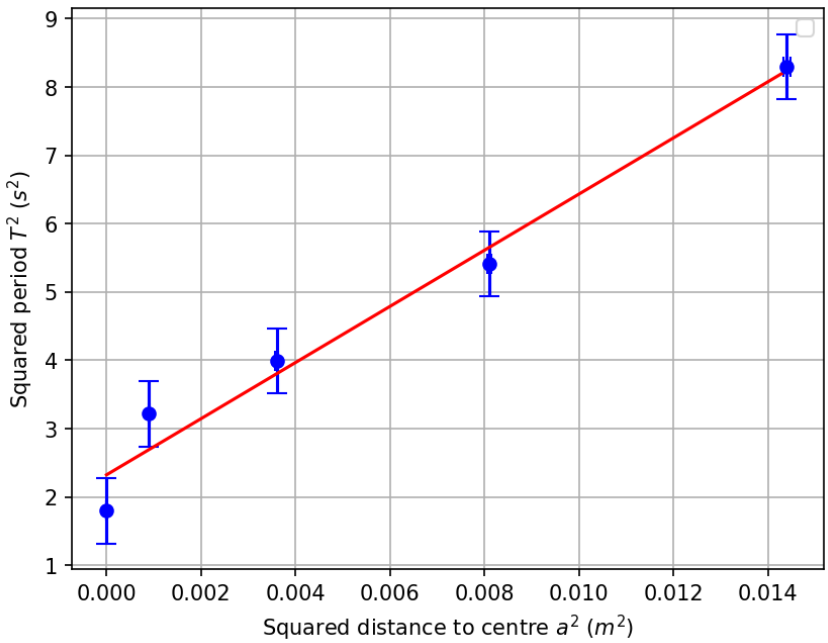


Figure 2: Linear regression of the square of the period (T^2) in function of the square of the distance (a^2).

This low discrepancy confirms that the dynamic behavior of the pendulum follows the theoretical prediction given by the term $\frac{4\pi^2 M}{D}$, thus validating the Parallel Axis Theorem for this setup.

References

[1] J.B. Marion. *Dinámica clásica de las partículas y sistemas*. Madrid: Reverté, S.A., 1989.

[2] Universidad de Granada. *Plataforma PRADO*. http://prado.ugr.es. Accessed: October 2025.